

13.5 Writing Algebraic Expressions

Lesson Introduction

 Benchmark	MA.6.AR.1.1 Given a mathematical or real-world context, translate written descriptions into algebraic expressions and translate algebraic expressions into written descriptions.		 Learning Targets	- I can translate a written description into an algebraic expression. - I can translate a real-world description into an algebraic expression.
 Benchmark Coherence	Building On MA.5.AR.2.1 Translate written real-world and mathematical descriptions into numerical expressions and numerical expressions into written mathematical descriptions.		Working Toward MA.7.AR.2.2 Write and solve two-step equations in one variable within a mathematical or real-world context, where all terms are rational numbers.	
 Essential Questions	- How do you write an algebraic expression from a written description? - How are expressions and equations similar? How are they different? - How can you use models to represent algebraic expressions? - How can algebraic expressions represent real-world situations?			
 Lesson Narrative	The lesson focuses on writing algebraic expressions for mathematical and real-world contexts. This is the first experience that students will have with writing expressions including a variable to represent an unknown value. The focus should be on understanding the action of the written description and selecting the appropriate mathematical operation based on that action. The use of bar models will help students make sense of the action and operation.			
 Component Summary	13.5.1 Warm-Up (~5 min.)	Matching and creating diagrams of numerical expressions (<i>SE p. 285</i>)	13.5.4 Guided Instruction (10 min.)	Write algebraic expressions based on real-world scenarios (<i>SE p. 289</i>)
	13.5.2 Guided Instruction (15 min.)	Represent algebraic expressions using bar models (<i>SE p. 285</i>)	13.5.5 Try It (5-10 min.)	Students write algebraic expressions to represent situations (<i>SE p. 290</i>)
	13.5.3 Collaboration (10 min.)	Match written descriptions to bar models and expressions with a partner (<i>SE p. 288</i>)	13.5.6 Wrap-Up (~5 min.)	Reflection (<i>SE p. 290</i>)
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> - Algebraic expression <u>Additional Terms:</u> - Equation		N/A	If allowing students to use rods or other manipulatives to form bar models, have those ready prior to the lesson.

 Teacher Tips	Common Misconceptions - Students may want to use key words to write the expressions and skip the use of the bar models. - Students may struggle with the order of the terms in subtraction expressions. - Students may struggle with the order of the dividend and the divisor in division expressions. - Students may struggle to differentiate between the parts and the whole in bar models.	Things to Consider - Although this lesson addresses writing expressions, not equations, it is important for students to think about the action versus key words. These models will carry over to help students write equations to match written descriptions and also to writing scenarios to match an expression or equation. - Bar models force students to think about the meaning of the words and the actions indicated. This will help students write the algebraic expression rather than strictly translating key words which often leads to having subtraction or division expressions written in the reverse order. - If time is an issue, consider assigning 13.5.5 Try It for homework.															
	Literacy and Language Routines MLR 7: Compare and Connect To foster students’ meta-awareness of different mathematical approaches and representations, teachers should demonstrate thinking out loud as they create and interpret bar models. Emphasize that bar models can be constructed in multiple ways to represent the same expression. Students should be prompted to reflect and respond as they discuss different approaches to creating bar models.	Mathematical Thinking and Reasoning (MTR) Standard MA.K12.MTR.4.1: Discussion and Reflection Students will be working to create and interpret bar models to represent expressions. When working with a partner, encourage students to clearly communicate their ideas and to collaborate with their partner to analyze each other’s ideas. Students will also justify their responses and recognize and correct errors in collaboration with their partner.															
	 Differentiate Instruction	Bar models drive the instruction throughout Lesson 5. Consider providing additional scratch paper and colored pencils or markers for students to engage with more practice as they work on each component in the lesson. Consider having groups construct their own anchor charts on bar models for additional scaffolding.	<table><tr><td>Equal Part</td><td>Equal Part</td><td>Equal Part</td><td>Equal Part</td><td>Equal Part</td><td>Equal Part</td><td>Equal Part</td><td>Equal Part</td></tr></table> Whole								Equal Part	Equal Part	Equal Part	Equal Part	Equal Part	Equal Part	Equal Part
Equal Part	Equal Part	Equal Part	Equal Part	Equal Part	Equal Part	Equal Part	Equal Part										
Lesson Notes:																	

13.5.1 Warm-Up: Bell Work

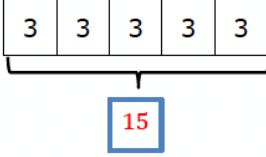
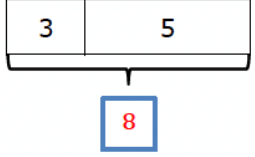
Component Overview: Students engage with visual models as they orient to the content of the lesson. Students begin by calculating the value of given diagrams before producing diagrams of their own for addition and multiplication problems.



13.5.1 Warm-Up: Bell Work

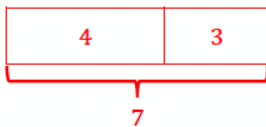
- Two diagrams are shown. One represents $3 + 5 = 8$. The other represents $5 \cdot 3 = 15$.

Complete the table by writing the equation that matches each diagram in the space provided and labeling the length of each diagram.

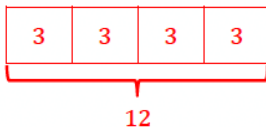
Equation	$5 \cdot 3 = 15$	$3 + 5 = 8$
Diagram		

- Draw a diagram that represents each equation.

a. $4 + 3 = 7$



b. $4 \cdot 3 = 12$



The diagram could also be constructed with three equal segments of 4.



Monitor for and highlight flexibility and fluency in student diagrams. For example, ask one student who drew four groups of three and another who drew three groups of four to display and discuss their equivalence.

13.5.2 Guided Instruction: Making Sense with Models

Component Overview: Students review the difference in algebraic expressions and equations. They then use bar diagrams to represent expressions. Encourage students to pay particular attention to how a bar model represents each expression. Focus on the action in the descriptions rather than memorizing the types of bar models.



13.5.2 Guided Instruction: Making Sense with Models

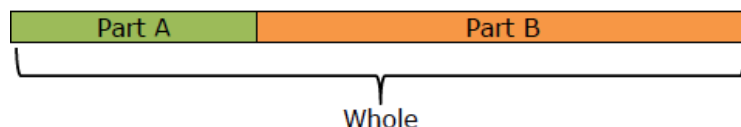
An **algebraic expression** is a mathematical statement containing numbers, operators, and at least one unknown, represented by a variable.

- Complete the statement.

An expression
☐ does
☒ does not
 have an equal sign or inequality symbol.

Bar models help connect the action of a problem to an algebraic expression.

Part-Part-Whole Model



The part-part-whole model can be used to show the following actions.

Action	Equation
Combining or joining	Part A + Part B = Whole
Taking away or separating	Whole - Part A = Part B Whole - Part B = Part A

- "The sum of a number and three" can be shown as the length of the whole bar where one part is n and the other part is 3.

The sum of a number and three



Write an algebraic expression for "The sum of a number and three."

$n + 3$



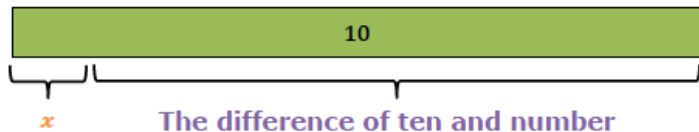
Position students to connect the visual models of the 13.5.1 Warm-Up to the bar model.



Prompt students to discuss the difference between identifying key words and focusing on the action of the story. Only identifying key words can lead to misconceptions; instead, focusing on the action produces more conceptual understanding of real-world problems.

13.5.2 Guided Instruction: Making Sense with Models (continued)

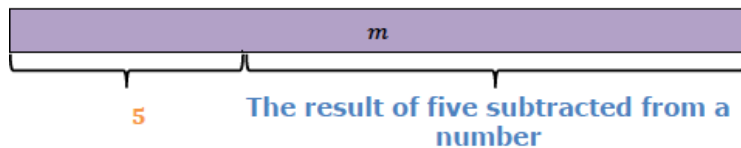
3. "The difference of ten and a number" can be shown where the whole bar is 10 and one part is a number represented by a variable, x . The difference is the other part.



Write an algebraic expression for "The difference of ten and a number."

$$10 - x$$

4. "Five subtracted from a number" is shown with the whole bar as a number represented by a variable, m , and one part is 5. The other part is the result of 5 being subtracted from the number, m .



Complete the statement.

The algebraic expression for "Five subtracted from a number, m " is $m - 5$.



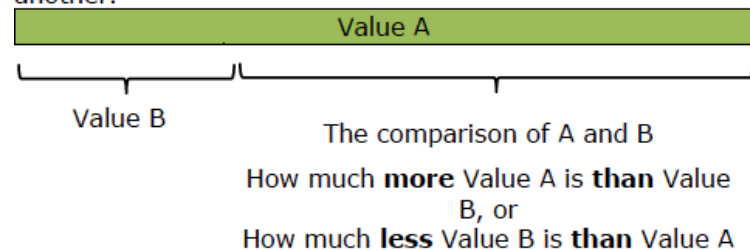
If students are struggling to connect the bar model to an expression, scaffold the instruction further by asking:

- What does "difference" mean?
- What is the whole? What are the parts?
- How can you label the whole on the bar model? How can you label the parts?
- What do you need to do with the parts to find the whole?

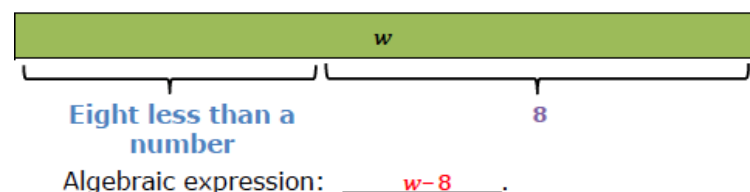
13.5.2 Guided Instruction: Making Sense with Models (continued)

Addition/Subtraction Comparison Model

Comparison models are much like part-part-whole models. They show how much more or less one value is than another.

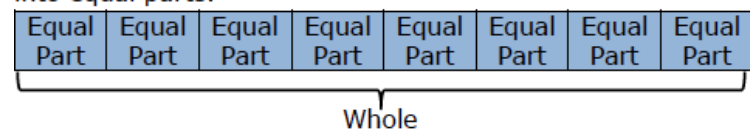


5. Write an algebraic expression for the bar model shown.



Equal Parts – Whole Model

The Equal Parts – Whole Model shows the whole sectioned into equal parts.



The equal parts-whole model can be used to show the following actions.

Action	Equation
Repeated addition (multiplication)	Equal part + ... + equal part = whole Equal part $\times$ number of parts = whole
Equal sharing	Whole $\div$ number of parts = equal part
Partitioning	Whole $\div$ equal part = number of parts



Students may struggle with the lack of detail in the description subtracted from a number. Rephrase to a question “What is five subtracted from your age?” Then, ask students if the answer would be same if you asked that question to your parents or guardian. Next, discuss where a variable should be used in the expression and why. Ask students how the whole could be represented. Finally, discuss the parts and how they could be represented.



Review the meanings of part and whole and use diagrams or images to explain the concepts. Allow students to use different lengths of rods to explore the concept of comparison.

Encourage students to verbalize their comparisons, such as, “The red rod is twice as long as the blue rod.”

13.5.3 Collaboration: Matching Activity

Component Overview: Students will collaborate to match bar models to written descriptions in addition to composing algebraic expressions to represent specific situations. Both activities review skills developed in the previous lesson.

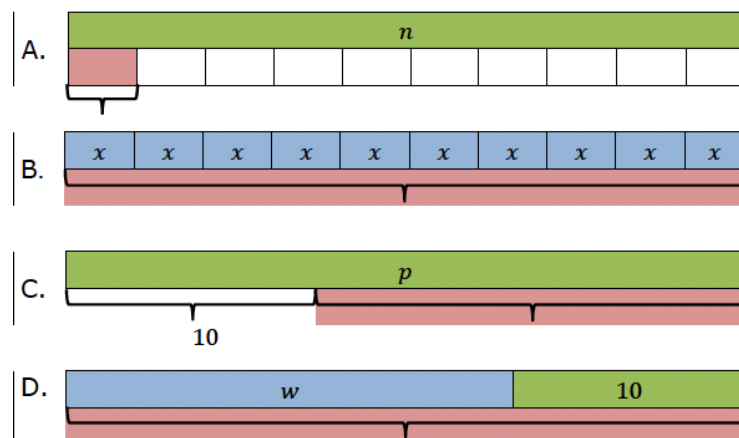


13.5.3 Collaboration: Matching Activity

Part 1

Match each written description to the appropriate bar model, where the red portion is the result of the written description. Write the letter of the bar model beside the corresponding word description.

Written Description	Bar Model
1. Ten subtracted from a number	<i>C</i>
2. A number divided by ten	<i>A</i>
3. The sum of a number and ten	<i>D</i>
4. A number times ten	<i>B</i>



Circulate as students work on 13.5.3 Collaboration, noting any gaps in learning or misconceptions that students have. Alternately, use information and observations from Lessons 1-4 to determine if small group intervention should be used during 13.5.3 Collaboration.



This section provides visual entry points through the use of bar models to represent each situation. Encourage students to think about and orally describe how A and B relate to each other to provide auditory entry points.



Activate prior knowledge to help students make connections between the written descriptions and bar models. Remind students that in a bar model, one bar represents the whole and the other bar represents the parts. Ask probing questions like:

- In C, what represents the whole? What are the parts?
- How do you know that A shows division?
- What algebraic expression could represent the bar model in B?
- What subtraction expression could D represent?

13.5.3 Collaboration: Matching Activity (continued)

Part 2

Turn and talk with your partner to compare your matches. Then, work with your partner to write an algebraic expression that represents the written description and bar model.

Written Description	Algebraic Expression
1. Ten subtracted from a number	$p - 10$
2. A number divided by ten	$\frac{n}{10}$
3. The sum of a number and ten	$w + 10$
4. A number times ten	$10x$



Students may struggle with the abstraction of variables. Consider giving students a relatable example to help them determine where the variable should go. Encourage students to substitute an actual number for the words “a number,” use that number to write the expression, and then replace the number with a variable.



Discuss how each written description is represented by an algebraic expression and a corresponding bar model. Encourage students to verbally explain the relationships seen within each grouping. For example, “10 subtracted from a number can be shown as the expression $p - 10$ because p represents the unknown number that 10 is subtracted from. In the bar model, the green bar represents the total that 10 is subtracted from. The pink bar shows the amount remaining.”

13.5.4 Guided Instruction: Writing Algebraic Expressions from Descriptions of Real-World Contexts

Component Overview: Students will write algebraic expressions to represent a given situation without using a concrete bar model. This encourages students to transition to thinking abstractly about algebraic expressions.



13.5.4 Guided Instruction: Writing Algebraic Expressions from Descriptions of Real-World Contexts

Algebraic expressions can be used to represent real-world situations.

1. Misty has 34 pairs of shoes arranged on a number of shelves. The same number of shoes are on each shelf. She has h number of shelves.
 - a. Write an algebraic expression that represents this situation.

$$\frac{34}{h}$$
 - b. What does the algebraic expression represent?
The expression represents the number of pairs of shoes on each shelf.
2. Martin buys 8 pounds of apples at a price per pound to make pie. Each pound costs p dollars.
 - a. Write an algebraic expression to represent this situation.
 $8p$
 - b. What does the algebraic expression represent?
The expression represents the total cost of the apples.
3. Paige is 6 inches taller than her brother. Paige's brother is m inches tall.
 - a. Write an algebraic expression to represent this situation.
 $m + 6$
 - b. What does the algebraic expression represent?
The expression represents Paige's height.



Three Reads

Implement a Three Reads routine here for each real-world situation.



Use the first read to help students understand context. Ask "What is this situation about?"

After the second read, ask students "What are the quantities in the situation?"

After the third read, ask students to brainstorm possible strategies to connect the situation with the appropriate expression.

Encourage students to repeat this routine themselves for each situation. This helps students connect the language in the question with the expression while keeping the intended level of cognitive demand.



If students are struggling to create algebraic expressions, allow them to use the middle step of first drawing a bar model to represent each situation. Also encourage students to verbally describe the action in each scenario before writing an algebraic expression for it. These modifications provide visual and auditory entry points.

13.5.5 Try It: Writing Algebraic Expressions

Component Overview: Students will practice what they have learned about creating algebraic expressions to represent specific situations. Remind students to look for the action instead of depending on key words when formulating an expression.



13.5.5 Try It: Writing Algebraic Expressions

Write an algebraic expression to represent each situation.

- Joe is 9 years younger than Drew, who is n years old.
 $n - 9$

- Jenna's salary, s , is lowered by \$120.
 $s - 120$

- Maria has 12 cupcakes and wants to split them among f friends.
 $\frac{12}{f}$

- Target has 8 cookie packages with c cookies in each.
 $8c$



Consider allowing students to work in pairs and then compare their expressions with those created by another student pair. Encourage students to verbally explain why they created each algebraic expression as they did and to describe which part of an expression matches which part of the written situation.

13.5.6 Wrap-Up: Reflection

Component Overview: Students will assess their understanding of creating an algebraic expression to represent a written description. Encourage students to thoroughly explain what they do and do not understand from the lesson.



13.5.6 Wrap-Up: Reflection

- Choose the emoji that reflects your understanding of translating a written description into an algebraic expression.

Answers will vary.



0 1 2 3 4 5 6 7 8 9 10

- Complete the statement.

I chose this emoji because _____

Answers will vary.



There is no right or wrong answer to the Reflection. Encourage students to thoughtfully assess their understanding of the lesson and to provide a concrete example as part of their explanation of why they chose that emoji.